Driver Behaviour Classification

Objective

Objective: The primary objective of this project is to develop a deep learning-based image classification model that can automatically detect and classify different driver behaviors. The system aims to distinguish between safe driving and various distracted behaviors such as texting, calling, or turning away from the road

The goal is to achieve high classification accuracy and reliable real-time performance.

Model: VGG16, MobileNet and AlexNet

Dataset

Data source: Kaggle
This image dataset contains the major activities of driver classified.

Dataset Includes 5 different class:

- Safe_driving
- Turning
- Other_activities
- Talking_phone
- texting_phone

Data Preparation

- Removed corrupted images.
- 70% Train, 10% Validation, 20% Test split.
- Used stratified sampling.

Stratification ensures that each split maintains the same class distribution. The test set was kept completely unseen during training and model selection.

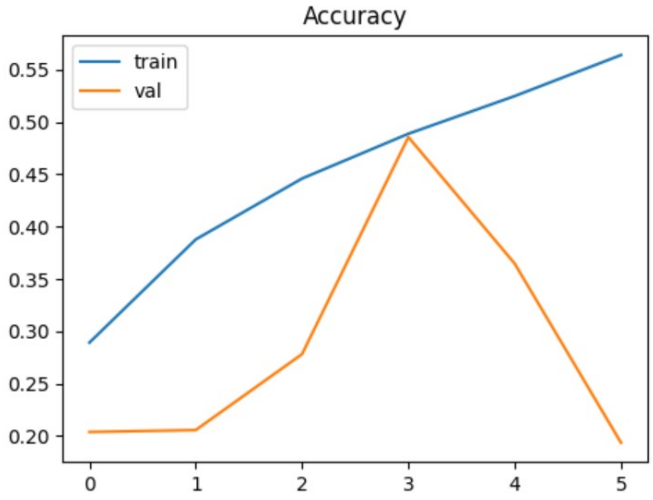
Baseline CNN Model

A simple CNN was implemented as a baseline model. This model was trained from scratch without any pretrained weights.

- 3 Convolution blocks.

  (Conv + BatchNorm + ReLU + MaxPooling)
- Global Average Pooling
- Dense layer with Dropout
- Softmax output layer

Images were normalized by scaling pixel values to the range [0,1].



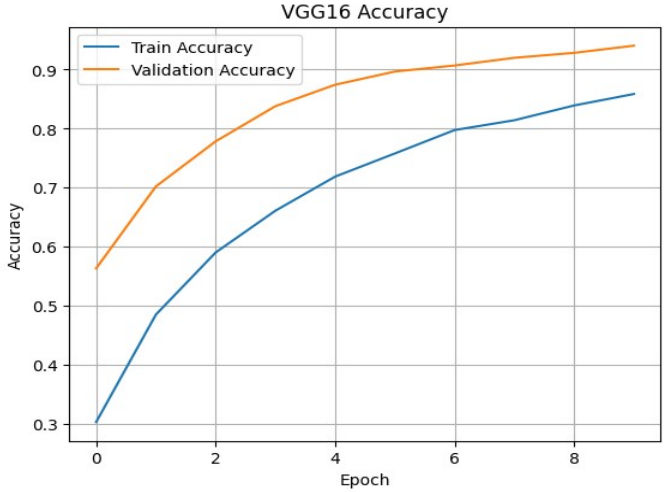

```
Baseline Test Accuracy: 0.4993

Baseline Test loss: 1.1752
```

VGG16 Transfer Learning

To improve performance, transfer learning was applied using the pretrained VGG16 model trained on ImageNet.

Two training phases were used:

1. Feature Extraction:
   - The convolutional base was frozen.
   - Only the custom classification head was trained.
2. Fine-Tuning:
   - The deeper convolutional layers were unfrozen.
   - A smaller learning rate was used.



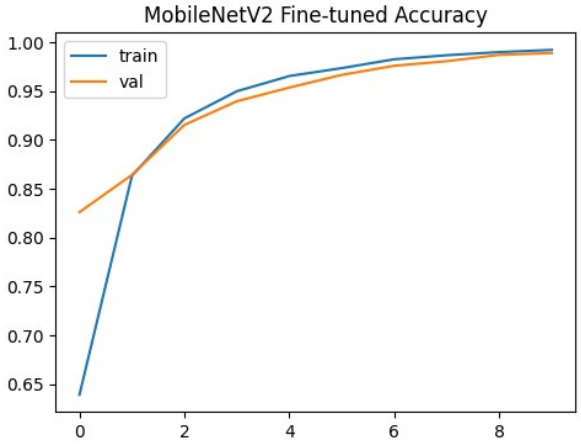

```
VGG16 Test Accuracy: 0.9335 and VGG16 Test Loss: 0.2753
VGG16 Fine-tuned Test Accuracy: 0.9926
VGG16 Fine-tuned Test Loss: 0.0350
```

MobileNetV2 Transfer Learning

MobileNetV2, a lightweight architecture optimized for efficiency, was also evaluated using transfer learning.

Similar to VGG16, the model was trained in two stages:

- Frozen feature extraction
- Fine-tuning with a lower learning rate



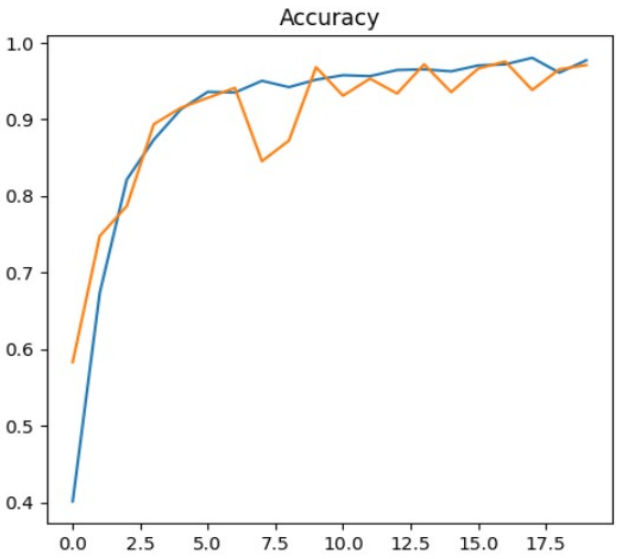

```
MobileNetV2 Test Accuracy: 0.8880

MobileNetV2 Fine-tuned Test Loss: 0.2993

MobileNetV2 Fine-tuned Test Accuracy: 0.9833

MobileNetV2 Fine-tuned Test Loss: 0.0669
```

AlexNet

It became famous for its ability to classify images accurately.

Architecture includes:

- 5 convolutional layers with Max-Pooling
- Max-Pooling uses a 3×3 filter with stride 2 which improved performance by reducing top-1 error by 0.4% and top-5 error by 0.3% compared to non-overlapping pooling.
- Followed by 2 fully connected layers each using dropout to prevent overfitting.
- Ends with a softmax layer for final classification.




```
Test Accuracy: 0.9237
Test Loss: 0.1794
```

Model Comparison

- The baseline CNN achieved moderate accuracy, showed signs of underfitting.
- **VGG16** – Deep architecture with strong feature extraction capability. Achieved the highest accuracy.
- **MobileNet** – Lightweight architecture, faster training, fewer parameters, efficient for real-time or mobile deployment.
- **AlexNet** achieved competitive accuracy. Larger number of parameters, slower training compared to MobileNet.

	Model	Test Accuracy	Test Loss
1	VGG16 (Fine-tuned)	0.9870	0.0443
2	MobileNetV2 (Fine-tuned)	0.9814	0.0672
3	AlexNet	0.9293	0.3069
0	Baseline CNN	0.4993	1.1737

Testing on New Image

VGG16 model prediction

```
Raw prediction:[[0.1812 0.1952 0.2195 0.2071 0.1967]]
Predicted Class: talking_phone
Confidence: 0.21958415
```

Mobilenet model prediction

```
Raw prediction: [[0.0347 0.0823 0.0036 0.0026 0.8766]]
Predicted Class: turning
Confidence: 0.8766074
```

Alex_model prediction

```
Raw prediction: [[0.1781  0.0444 0.0214 0.0101 0.7458]]
Predicted Class: turning
Confidence: 0.7458455
```

Conclusion

- Developed a deep learning model to classify driver behavior from images.
- Model learned meaningful patterns but showed some overfitting on validation data.
- Transfer learning improved performance and generalization.
- The system shows strong potential for real-time driver monitoring and road safety applications.
- Testing on completely new external images revealed some performance variation. This highlights the importance of dataset diversity and real-world variability in improving generalization.
- Future improvements may include stronger data augmentation and expanding the dataset with more diverse driving conditions.

Thank You